

Lect2

Objectives:

- 1- Know what is resting membrane potential. Understand how different ions participate in action potential, phases of action potential and properties of action potential

Resting membrane potential

The voltage difference across the cell membrane of a neuron is called a **membrane potential**; it is the difference between the electrical potential in the cytoplasm of the cell and the electrical potential in the extracellular space .Figure1.

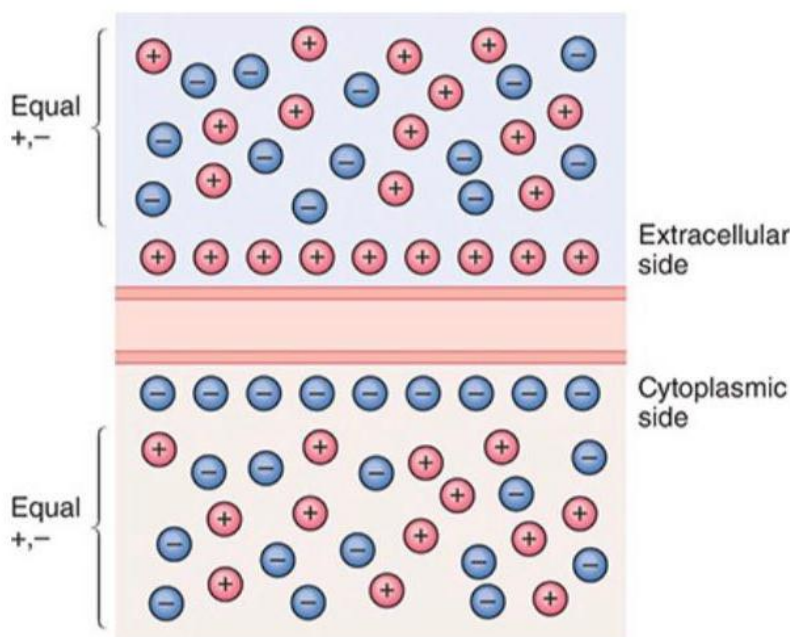


Figure1 membrane potential results from separation of positive and negative charges across the cell membrane.

A cell at rest has a potential difference across its membrane such that the inside of the cell is negatively charged relative to the outside. This difference is called the **resting membrane potential (RMP)** because the cell is at rest—it is not receiving or transmitting any signals. The resting

membrane potential of neurons is approximately -70 mV. That is, the potential *inside the fiber* is 70 millivolts more negative than the potential in the extracellular fluid on the outside of the fiber. This **resting membrane potential** is due to different distribution of ions inside and outside the membrane.table1

Ion	Intracellular	Extracellular
Na⁺	10 mEq/L	142 mEq/L
K⁺	140 mEq/L	4 mEq/L
Cl⁻	4 mEq/L	103 mEq/L
Phosphates	75 mEq/L	4 mEq/L
Proteins	200 mg/dL	30 mg/dL

Causes of resting membrane potential:

1.Active Transport of Sodium and Potassium Ions Through the Membrane—the Sodium-Potassium (Na⁺-K⁺)Pump.

all cell membranes of the body have a powerful Na⁺-K⁺ pump that continually transports sodium ions to the outside of the cell and potassium ions to the inside, by this *pump* three Na⁺ ions are pumped to the outside for each two K⁺ ions to the inside, leaving a net deficit of positive ions on the inside and causing a negative potential inside the cell membrane. The Na⁺-K⁺ pump also causes large concentration gradients for sodium and potassium across the resting nerve membrane.

2. Continuous passive **diffusion of K^+** outside the cell. This **diffusion potential** is the major factor responsible for RMP.

The outward movement of potassium exceeds the inward movement of sodium: potassium and sodium are able to cross the cell membrane, these ions move in the direction of their chemical driving forces—that is, potassium ions move out of the cell and sodium ions move into the cell. Nevertheless, the outward movement of potassium exceeds the inward movement of sodium because the permeability of the membrane to potassium is greater than the permeability to sodium. Under these conditions, a net outward movement of positive charge occurs, which gives rise to a negative membrane potential. Figure2

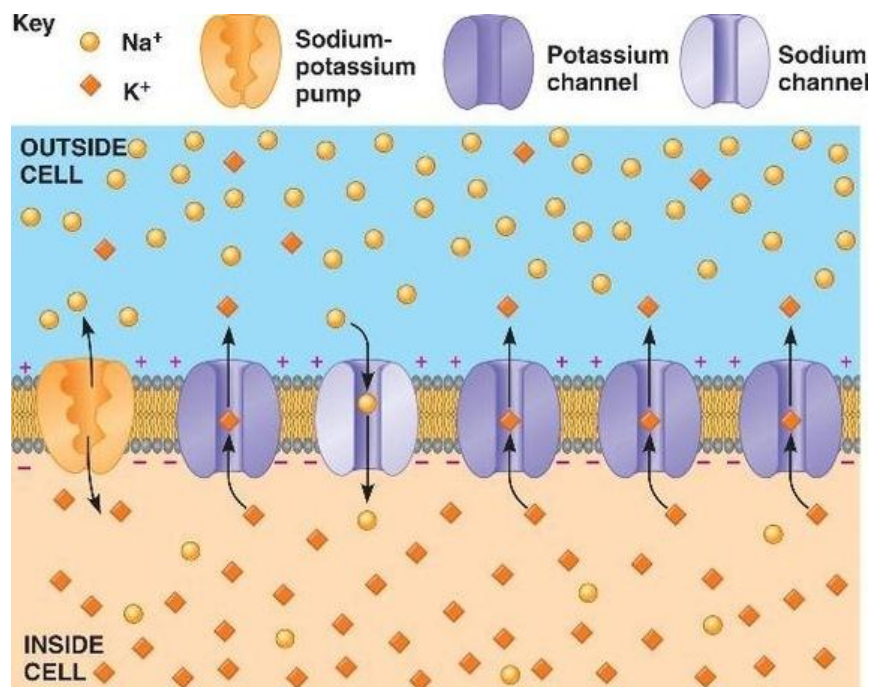


Figure2 Ions channels and resting membrane potential

Graded potentials are small changes in membrane potential that occur when ion channels open or close in response to a stimulus acting on the

cell. They occur at dendrites and soma of neurons due to gated ion channels opening in response to stimuli like neurotransmitters

Action potential :

Nerve and muscle cells are excitable cells. Action potentials occur in the membranes of excitable tissue (nerve and muscle) in response to graded potentials that reach threshold.

When a sufficient external **stimulus** is applied on their plasma membrane, the negativity is reversed and the internal side will become positively charged. The external stimulus may be electrical, chemical, physical or other types of stimuli. This change in membrane potential is called **action potential**. The response in nerve cell is **transmission** of action potential (nerve **impulse**) while the response in muscle cell is **contraction**.

An action potential in a neuron consists of three distinct phases:

1. Rapid Depolarization. The first phase of an action potential is a rapid depolarization during which the membrane potential changes from -70 mV (rest) to +30 mV. This depolarization is caused by a sudden and dramatic increase in permeability to sodium followed by an increase in the movement of sodium ions into the cell, down sodium's electrochemical gradient.
2. Repolarization. The second phase of an action potential is a repolarization of the membrane potential during which the membrane potential returns from +30 mV back to resting levels (-70 mV). Within 1 msec after the increase in sodium permeability, sodium permeability decreases rapidly, reducing the inflow of sodium. At approximately the same time, potassium permeability increases. Potassium then moves down its electrochemical

gradient out of the cell, repolarizing the membrane potential to bring it back to resting levels.

3. After-hyperpolarization. The third phase of an action potential is termed after-hyperpolarization. Potassium permeability remains elevated for a brief time (5–15 msec) after the membrane potential reaches the resting membrane potential, resulting in an after-hyperpolarization. Figure3

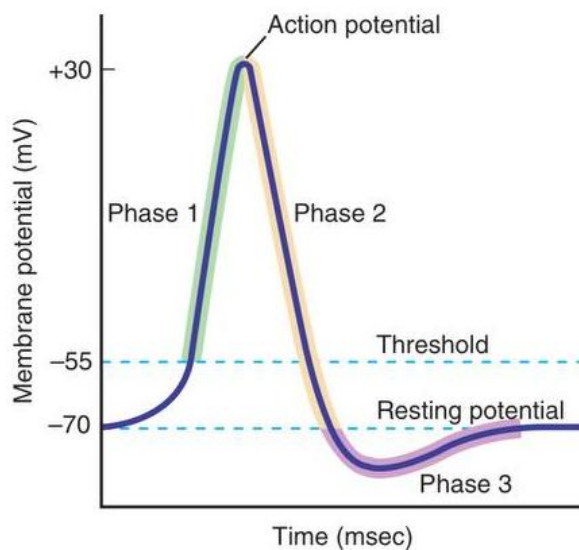


Figure3 Phases of action potential

Properties of action potential : the main properties of the action potential:

- 1- **Threshold intensity**: The membrane potential at which action potential starts is called **threshold potential** or **firing potential**. This critical level of depolarization must be reached at the **axon initial segment (axon hillock/trigger zone)** in order to initiate an **action potential (AP)** .

Threshold stimuli: When a sufficient stimulus raises the membrane potential 15 mV above RMP (i.e. from -70 mV to -55 mV); action

potential phases will start and does not stop until complete cycle occurs.

To reach threshold: **Graded potentials** will act to depolarize the membrane mainly at the axon hillock where voltage gated sodium channels are concentrated and as depolarization reaches -55 mV, **activation gates** of Na^+ channels start to open rapidly which causes further depolarization → positive feedback and then action potential begins. Voltage gated sodium channels at axon hillock are different from those distributed along the axon in that they have higher kinetics making them more suitable for action potential initiation.

-Subthreshold stimuli: Any weak stimuli that does not cause action potential is called sub threshold. they may cause **local responses** but they don't cause action potential. Local responses may increase or decrease according to the strength of stimulus and they subside after removal of these stimuli. Local responses stay in their location and don't travel (no propagation).

- **Supramaximal** stimuli induce the same effects as threshold stimuli and do not change the shape of action potential curve. Figure 4

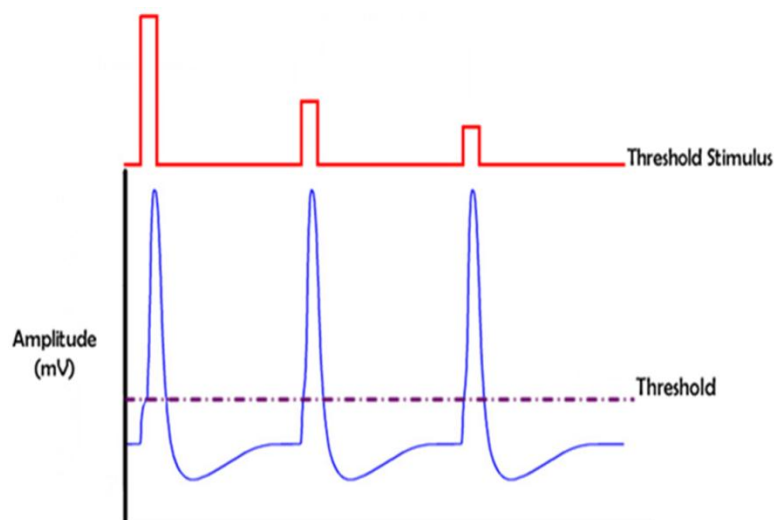
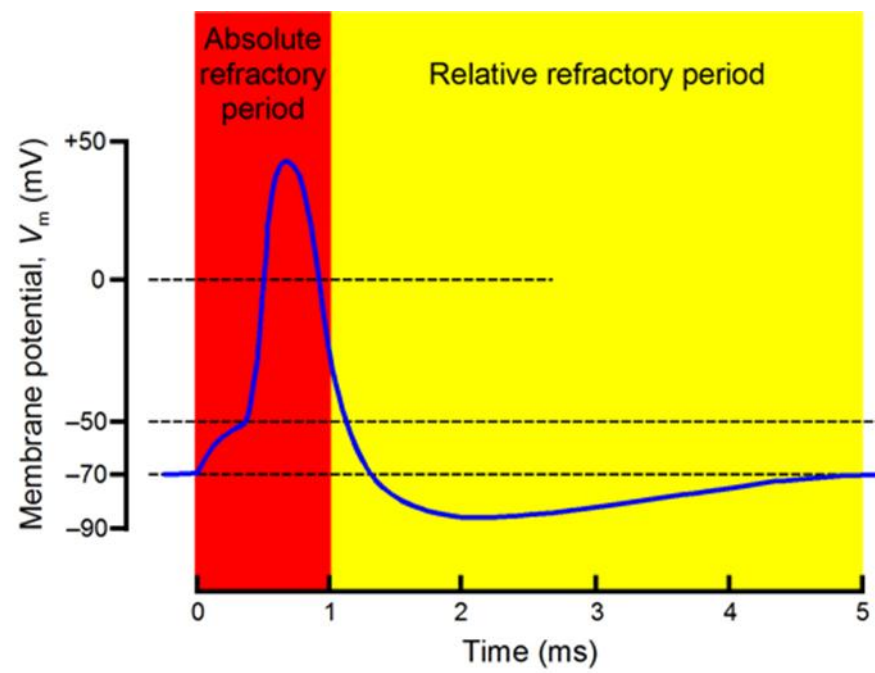


Figure4 threshold intensity

2-All or none principle : Initiation of action potentials follows the **all-or-none principle**: Whether a membrane is depolarized to threshold or greater, the amplitude of the resulting action potential is the same; if the membrane is not depolarized to the threshold, no action potential occurs.

3.Refractory Periods

During and immediately after an action potential, the membrane is less excitable than it is at rest. This period of reduced excitability is called the **refractory period**. The refractory period can be divided into two phases: the absolute refractory period and the relative refractory period .The **absolute refractory period** spans all of the depolarization phase plus most of the repolarization phase of an action potential (until 2/3 repolarization is complete). During this time, a second action potential cannot be generated in response to a second stimulus, regardless of the strength of that stimulus. The **relative refractory period** occurs immediately after the absolute refractory period (From the point of 2/3 repolarization to the end of repolarization phase) . During this period, it is possible to generate a second action potential, but only in response to a stimulus stronger than that needed to reach threshold under resting conditions. Figure4



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Figure4 Refractory periods